

Step 1 Create a GitHub Repository

1 General

Owner * Abrshi / Repository name *

✔ Your new repository will be created as `-tms-client`. The repository name can only contain ASCII letters, digits, and the characters `-,` and `.`

Great repository names are short and memorable. How about [refactored-octo-enigma](#)?

Description

49 / 350 characters

2 Configuration

Choose visibility * Public

Choose who can see and commit to this repository

Start with a template No template

Templates pre-configure your repository with files.

Add README Off

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore

.gitignore tells git which files not to track. [About .gitignore files](#)

Add license No license

Licenses explain how others can use your code. [About licenses](#)

Create repository

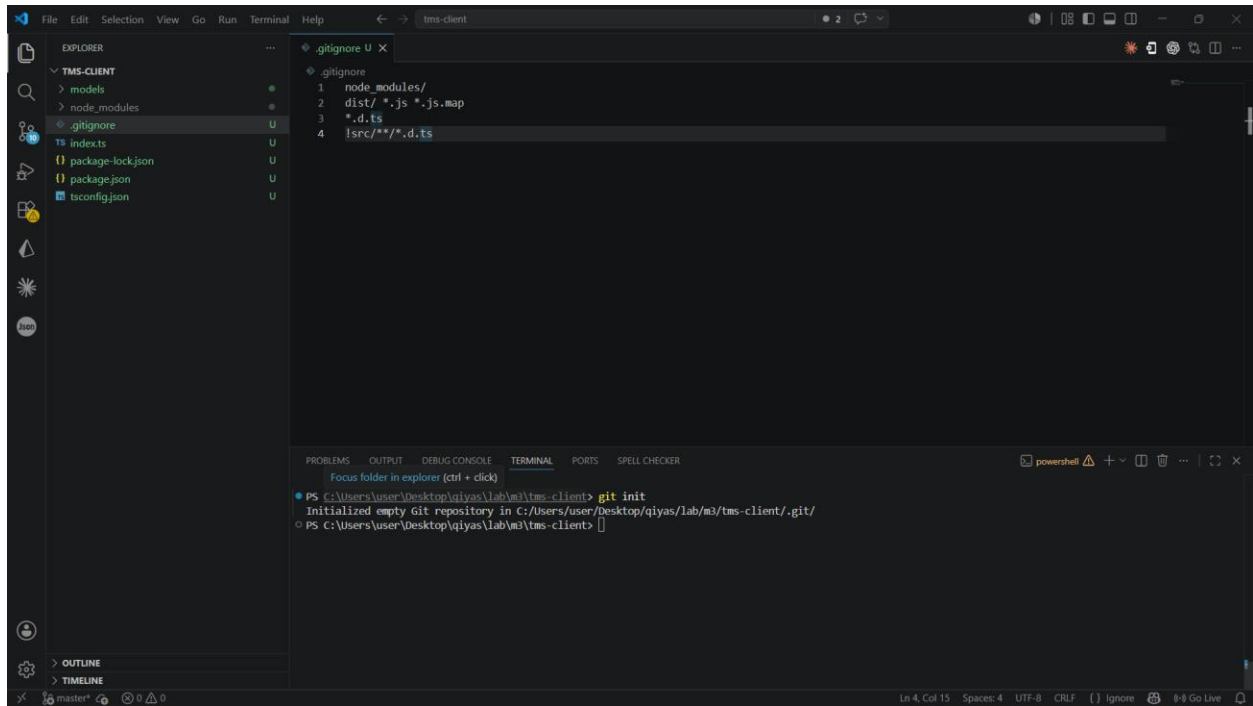
Step 2 Initialize Git in Your M2 Project

```
1 import { Temporal } from "@js-temporal/polyfill";
2 import { Student, isStudent, parseStudent } from "../models/student.model";
3 const student: Student = {
4   id: "STU-001",
5   name: "Hana Tadesse",
6   enrollmentDate: Temporal.Now.instant(),
7 };
8
9
10
11 // Try these what does the compiler say?
12 // student.id = "STU-999";
13 // console.log(student.gpa.toFixed(2));
14 console.log(student.gpa?.toFixed(2) ?? "Not yet graded");
15
16 // function processStudent(data: any) {
17 //   console.log(`GPA: ${data.gpa.toFixed(2)}`); // Crashes if gpa is missing or not a number
18 // }
19
20 function processStudent(raw: unknown) {
21   if (isStudent(raw)) {
22     const gpaDisplay = raw.gpa?.toFixed(2) ?? "Not yet graded";
23     console.log(`Student ${raw.name} GPA: ${gpaDisplay}`);
24   } else {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git init
Initialized empty Git repository in C:/Users/user/Desktop/qiyas/lab/m3/tms-client/.git/
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Step 3 Create a .gitignore



The screenshot shows the Visual Studio Code interface with the Explorer view on the left displaying the project structure. The file explorer shows the following files and folders:

- TMS-CLIENT
 - models
 - node_modules
 - .gitignore
 - index.ts
 - package-lock.json
 - package.json
 - tsconfig.json

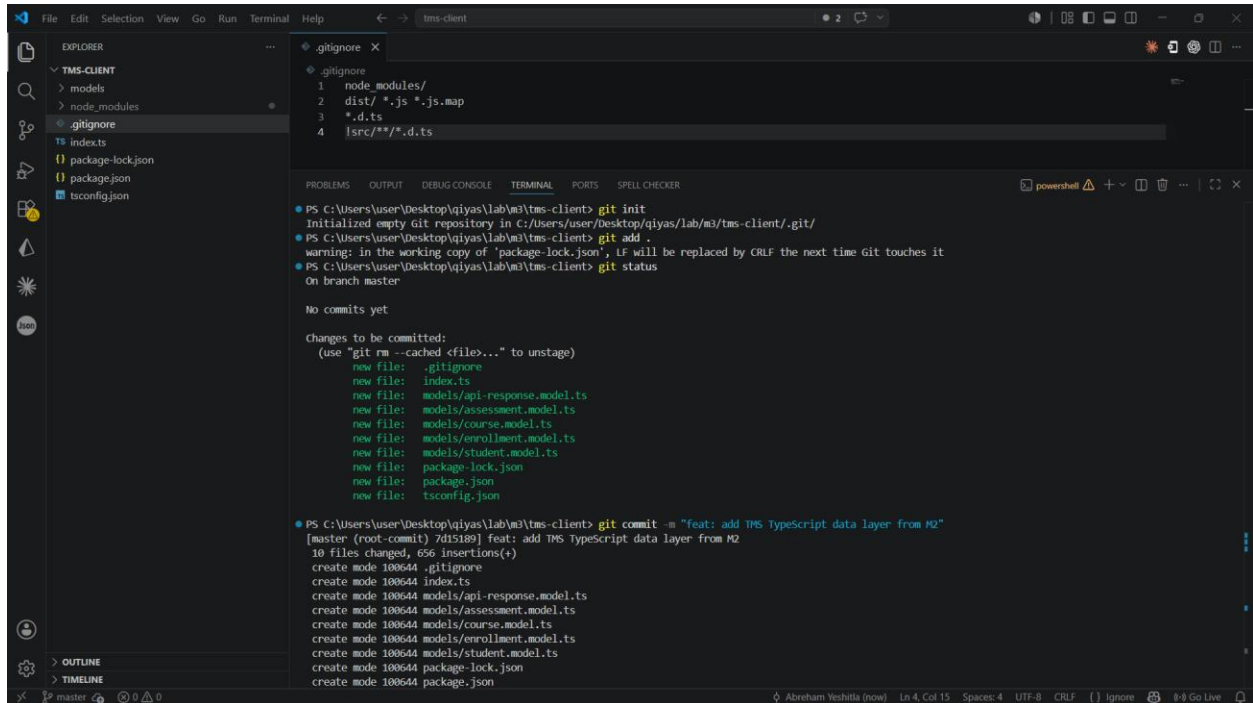
The main editor window shows the content of the .gitignore file:

```
.gitignore
1 node_modules/
2 dist/ *.js *.js.map
3 *.d.ts
4 !src/**/*.d.ts
```

The terminal window at the bottom shows the execution of the following commands:

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git init
Initialized empty Git repository in C:\Users\user\Desktop\qiyas\lab\m3\tms-client\.git\
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Step 4 Make Your First Commit



The screenshot shows the Visual Studio Code interface with the Explorer view on the left displaying the project structure. The file explorer shows the following files and folders:

- TMS-CLIENT
 - models
 - node_modules
 - .gitignore
 - index.ts
 - package-lock.json
 - package.json
 - tsconfig.json

The main editor window shows the content of the .gitignore file:

```
.gitignore
1 node_modules/
2 dist/ *.js *.js.map
3 *.d.ts
4 !src/**/*.d.ts
```

The terminal window at the bottom shows the execution of the following commands:

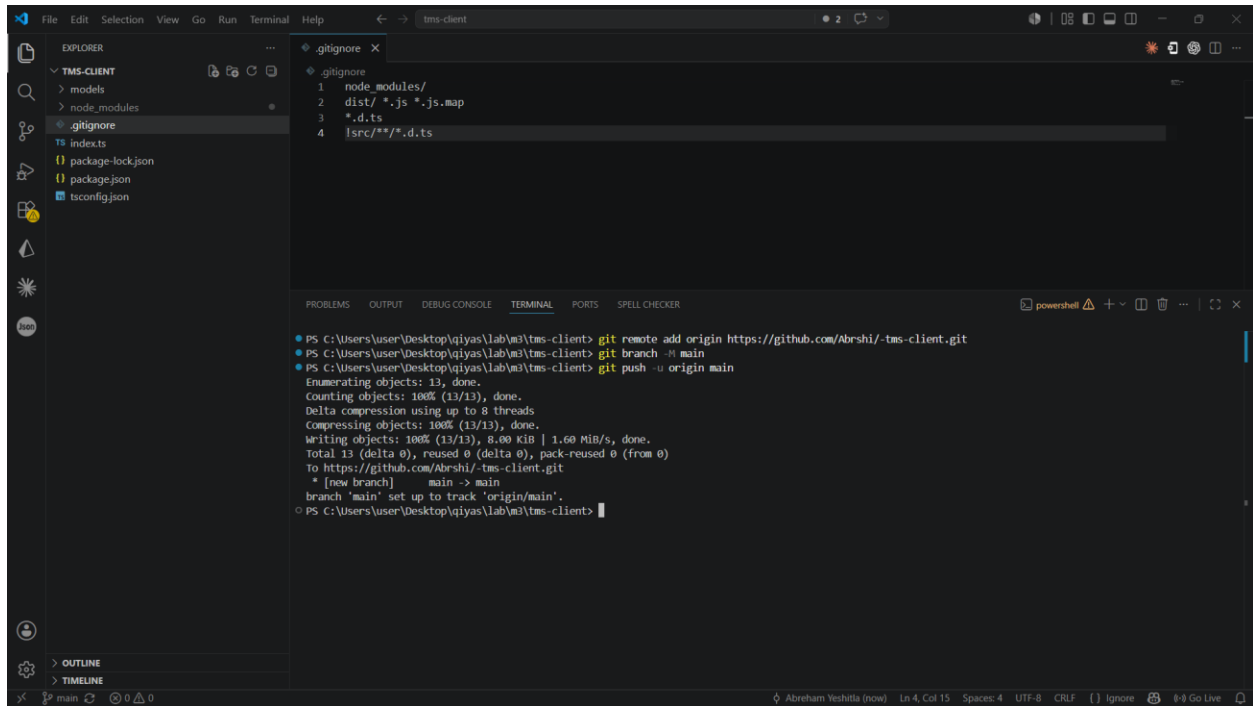
```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git init
Initialized empty Git repository in C:\Users\user\Desktop\qiyas\lab\m3\tms-client\.git\
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git add .
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   index.ts
    new file:   models/api-response.model.ts
    new file:   models/assessment.model.ts
    new file:   models/course.model.ts
    new file:   models/enrollment.model.ts
    new file:   models/student.model.ts
    new file:   package-lock.json
    new file:   package.json
    new file:   tsconfig.json

PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit -m "feat: add TMS TypeScript data layer from M2"
[master (root-commit) 7d15189] feat: add TMS TypeScript data layer from M2
10 files changed, 656 insertions(+)
create mode 100644 .gitignore
create mode 100644 index.ts
create mode 100644 models/api-response.model.ts
create mode 100644 models/assessment.model.ts
create mode 100644 models/course.model.ts
create mode 100644 models/enrollment.model.ts
create mode 100644 models/student.model.ts
create mode 100644 package-lock.json
create mode 100644 package.json
```

Step 5 Connect to GitHub and Push



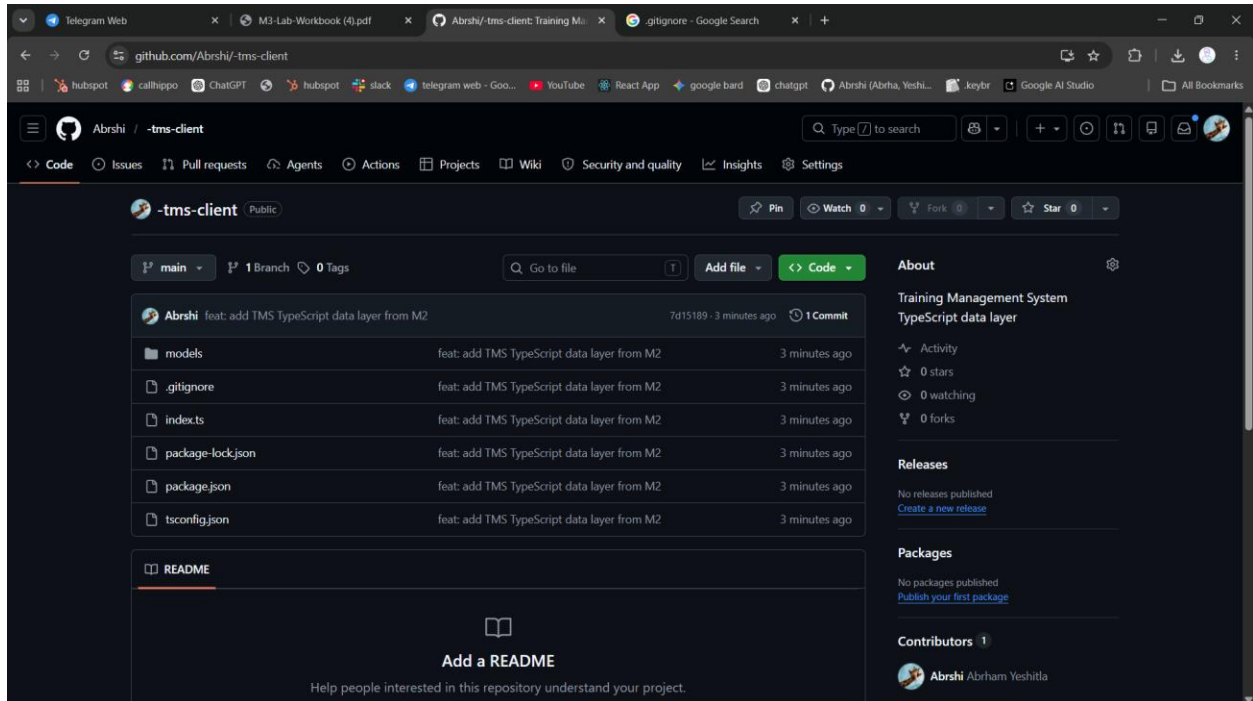
The screenshot shows the Visual Studio Code interface. The Explorer pane on the left shows the file structure of the `tms-client` project, including `models`, `node_modules`, `.gitignore`, `index.ts`, `package-lock.json`, `package.json`, and `tsconfig.json`. The `.gitignore` file is open in the editor, showing the following content:

```
.gitignore
1 node_modules/
2 dist/ *.js *.js.map
3 *.d.ts
4 !src/**/*.d.ts
```

The Terminal pane at the bottom shows the execution of the following commands in a PowerShell session:

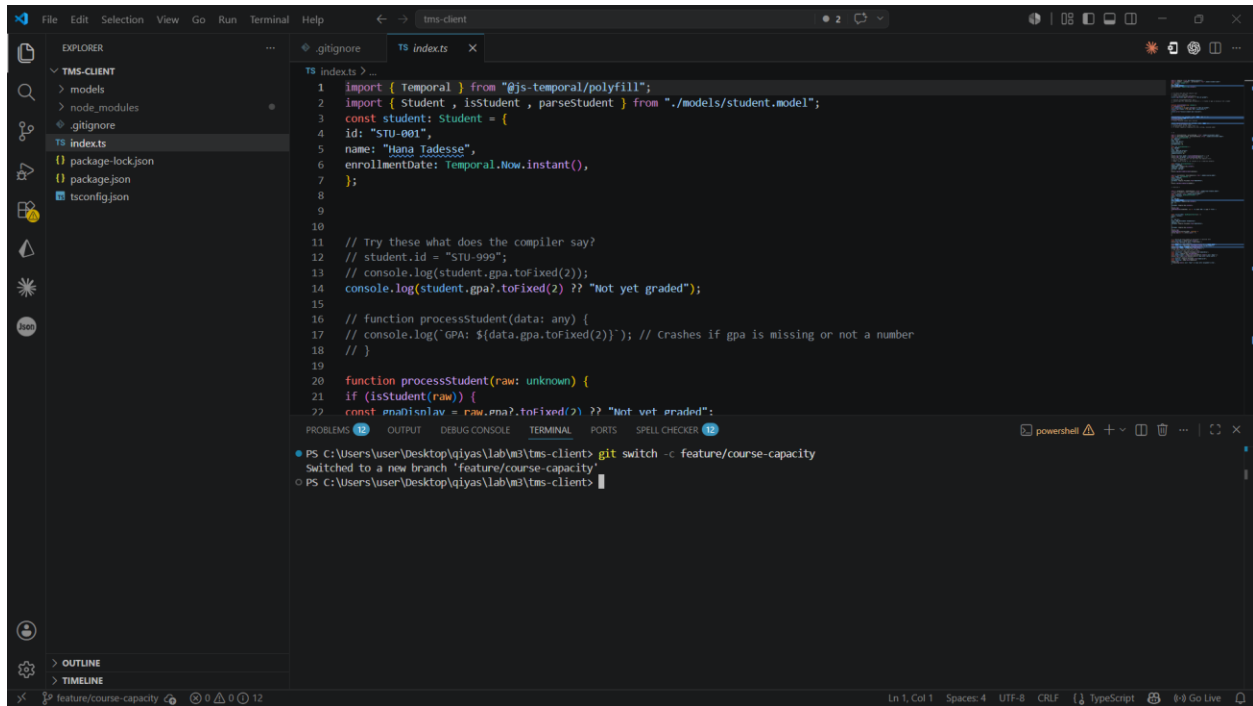
```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git remote add origin https://github.com/Abbrshi/-tms-client.git
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git branch -M main
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git push -u origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (13/13), 8.00 KiB | 1.60 MiB/s, done.
Total 13 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Abbrshi/-tms-client.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Step 6 Verify on GitHub



Exercise 1: Modern Git Syntax (switch and restore)

Step 1 Create a Feature Branch

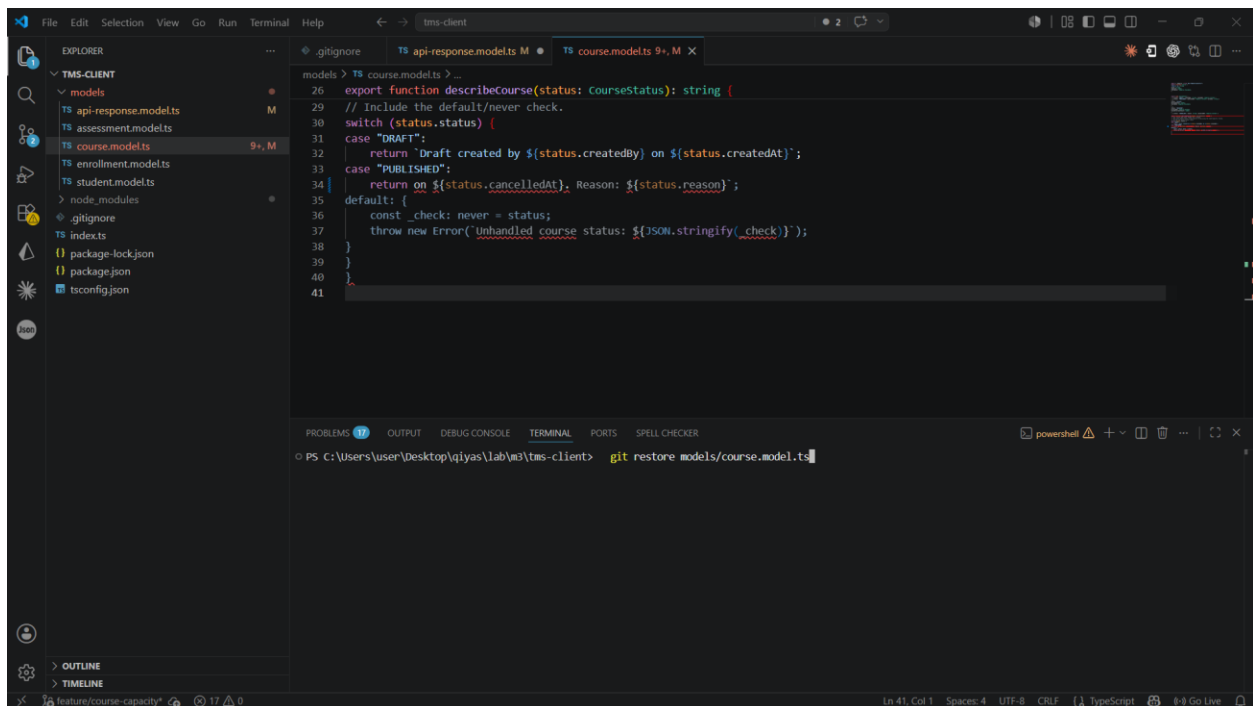


The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the file structure of the 'TMS-CLIENT' project. The main editor window is open to 'ts/index.ts'. The code in the editor includes imports for 'Temporal' and 'Student', a constant 'student' object, and a function 'processStudent'. The terminal at the bottom shows the execution of the command 'git switch -c feature/course-capacity', which successfully switches to a new branch named 'feature/course-capacity'.

```
1 import { Temporal } from "@js-temporal/polyfill";
2 import { Student, isStudent, parseStudent } from "../models/student.model";
3 const student: Student = {
4   id: "STU-001",
5   name: "Hana Tadesse",
6   enrollmentDate: Temporal.Now.instant(),
7 };
8
9
10
11 // Try these what does the compiler say?
12 // student.id = "STU-999";
13 // console.log(student.gpa.toFixed(2));
14 console.log(student.gpa?.toFixed(2) ?? "Not yet graded");
15
16 // function processStudent(data: any) {
17 //   console.log(`GPA: ${data.gpa.toFixed(2)}`); // Crashes if gpa is missing or not a number
18 // }
19
20 function processStudent(raw: unknown) {
21   if (isStudent(raw)) {
22     const gpaDisplay = raw.gpa?.toFixed(2) ?? "Not yet graded";
23   }
24 }
```

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch -c feature/course-capacity
Switched to a new branch 'feature/course-capacity'
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Step 2 Restore a File

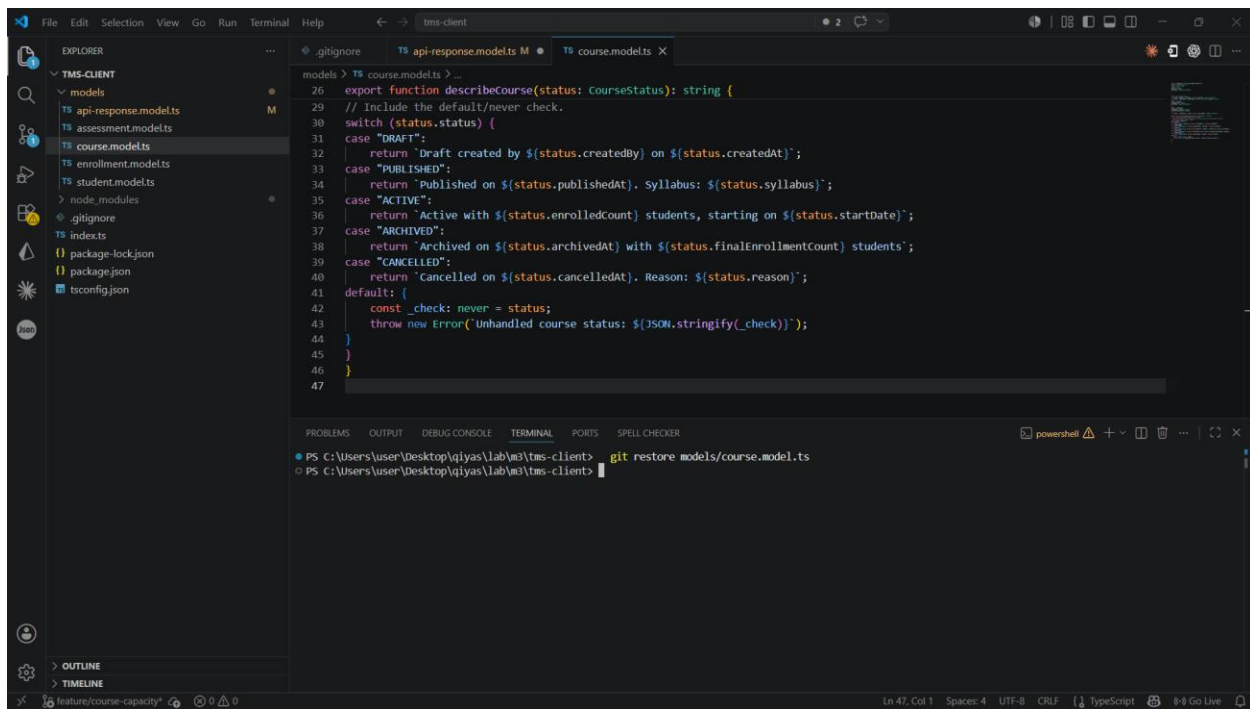


The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left. The main editor window is open to 'models/course.model.ts'. The code in the editor includes a function 'describeCourse' that uses a switch statement to handle different course statuses. The terminal at the bottom shows the execution of the command 'git restore models/course.model.ts', which restores the file to its original state.

```
26 export function describeCourse(status: CourseStatus): string {
27   // Include the default/never check.
28   switch (status.status) {
29     case "DRAFT":
30       return `Draft created by ${status.createdBy} on ${status.createdAt}`;
31     case "PUBLISHED":
32       return `on ${status.cancelledAt}, Reason: ${status.reason}`;
33     default:
34       const check: never = status;
35       throw new Error(`Unhandled course status: ${JSON.stringify(check)}`);
36   }
37 }
38
39
40
41
```

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git restore models/course.model.ts
```

After



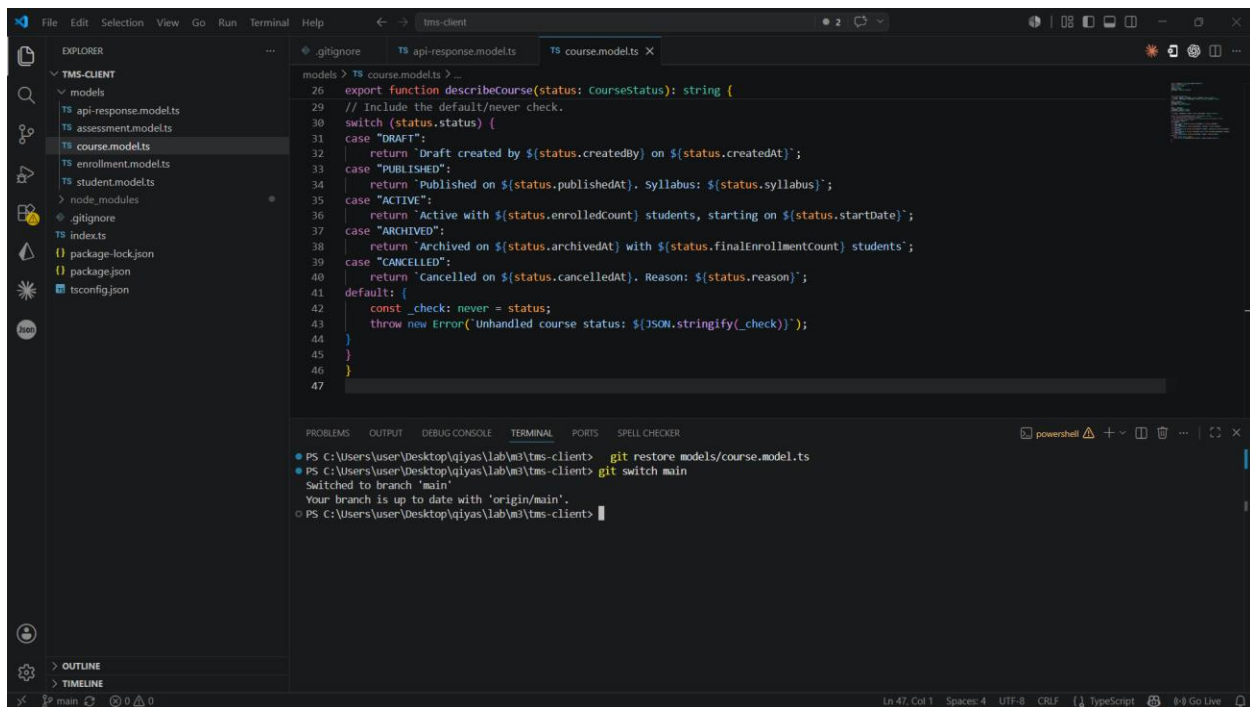
```
File Edit Selection View Go Run Terminal Help
tms-client

EXPLORER
TMS-CLIENT
  models
  api-response.models.ts
  assessment.models.ts
  course.models.ts
  enrollment.models.ts
  student.models.ts
  node_modules
  .gitignore
  index.ts
  package-lock.json
  package.json
  tsconfig.json

models > TS course.model.ts > ...
26 export function describeCourse(status: CourseStatus): string {
29 // Include the default/never check.
30 switch (status.status) {
31   case "DRAFT":
32     return `Draft created by ${status.createdBy} on ${status.createdAt}`;
33   case "PUBLISHED":
34     return `Published on ${status.publishedAt}, syllabus: ${status.syllabus}`;
35   case "ACTIVE":
36     return `Active with ${status.enrolledCount} students, starting on ${status.startDate}`;
37   case "ARCHIVED":
38     return `Archived on ${status.archivedAt} with ${status.finalEnrollmentCount} students`;
39   case "CANCELLED":
40     return `Cancelled on ${status.cancelledAt}. Reason: ${status.reason}`;
41   default: {
42     const _check: never = status;
43     throw new Error(`Unhandled course status: ${JSON.stringify(_check)}`);
44   }
45 }
46 }
47

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER
powershell
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git restore models/course.model.ts
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Step 3 Switch Back



```
File Edit Selection View Go Run Terminal Help
tms-client

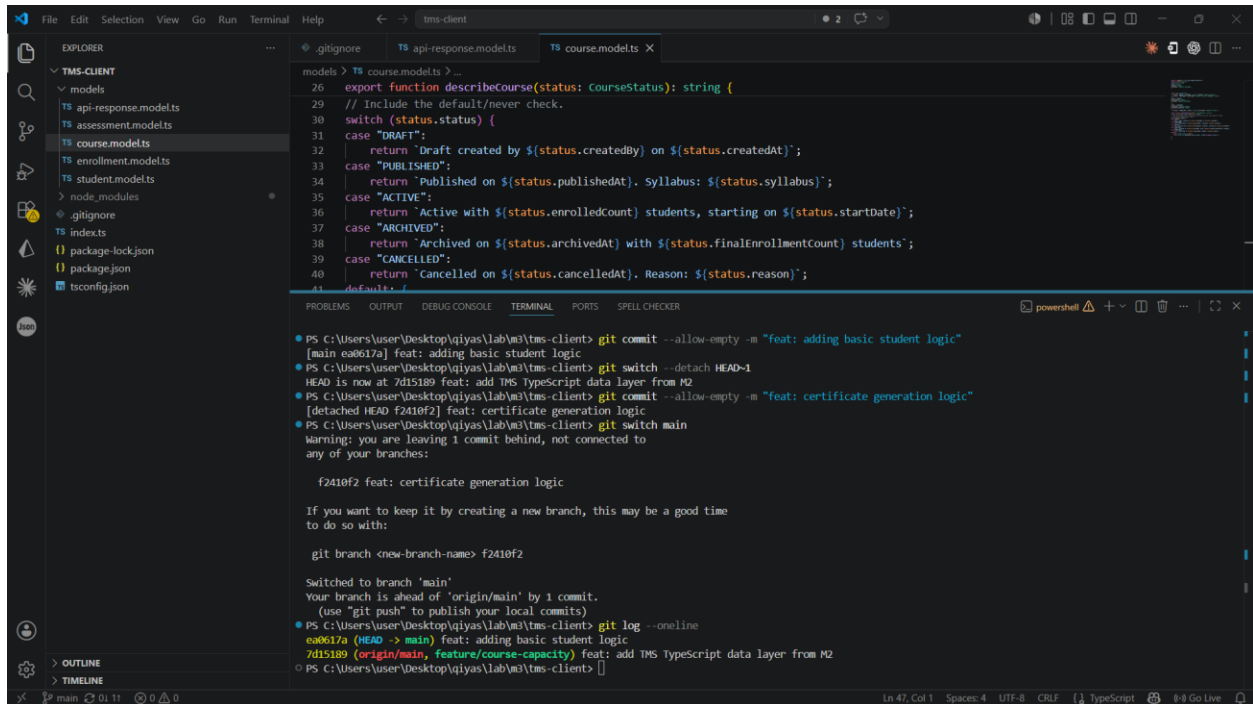
EXPLORER
TMS-CLIENT
  models
  api-response.models.ts
  assessment.models.ts
  course.models.ts
  enrollment.models.ts
  student.models.ts
  node_modules
  .gitignore
  index.ts
  package-lock.json
  package.json
  tsconfig.json

models > TS course.model.ts > ...
26 export function describeCourse(status: CourseStatus): string {
29 // Include the default/never check.
30 switch (status.status) {
31   case "DRAFT":
32     return `Draft created by ${status.createdBy} on ${status.createdAt}`;
33   case "PUBLISHED":
34     return `Published on ${status.publishedAt}, syllabus: ${status.syllabus}`;
35   case "ACTIVE":
36     return `Active with ${status.enrolledCount} students, starting on ${status.startDate}`;
37   case "ARCHIVED":
38     return `Archived on ${status.archivedAt} with ${status.finalEnrollmentCount} students`;
39   case "CANCELLED":
40     return `Cancelled on ${status.cancelledAt}. Reason: ${status.reason}`;
41   default: {
42     const _check: never = status;
43     throw new Error(`Unhandled course status: ${JSON.stringify(_check)}`);
44   }
45 }
46 }
47

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER
powershell
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git restore models/course.model.ts
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Exercise 2: The Timeline Panic

Step 1 Simulate the Disaster



```
models > TS course.models.ts > ...
26 export function describeCourse(status: CourseStatus): string {
29 // Include the default/never check.
30 switch (status.status) {
31 case "DRAFT":
32   return `Draft created by ${status.createdBy} on ${status.createdAt}`;
33 case "PUBLISHED":
34   return `Published on ${status.publishedAt}. Syllabus: ${status.syllabus}`;
35 case "ACTIVE":
36   return `Active with ${status.enrolledCount} students, starting on ${status.startDate}`;
37 case "ARCHIVED":
38   return `Archived on ${status.archivedAt} with ${status.finalEnrollmentCount} students`;
39 case "CANCELLED":
40   return `Cancelled on ${status.cancelledAt}. Reason: ${status.reason}`;
41 default:
42   return "Unknown status";
43 }
44 }
```

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit --allow-empty -m "feat: adding basic student logic"
[main ea617a] feat: adding basic student logic
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch --detach HEAD-1
HEAD is now at 7d15189 feat: add TMS TypeScript data layer from M2
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit --allow-empty -m "feat: certificate generation logic"
[detached HEAD f2410f2] feat: certificate generation logic
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch main
Warning: you are leaving 1 commit behind, not connected to
any of your branches:

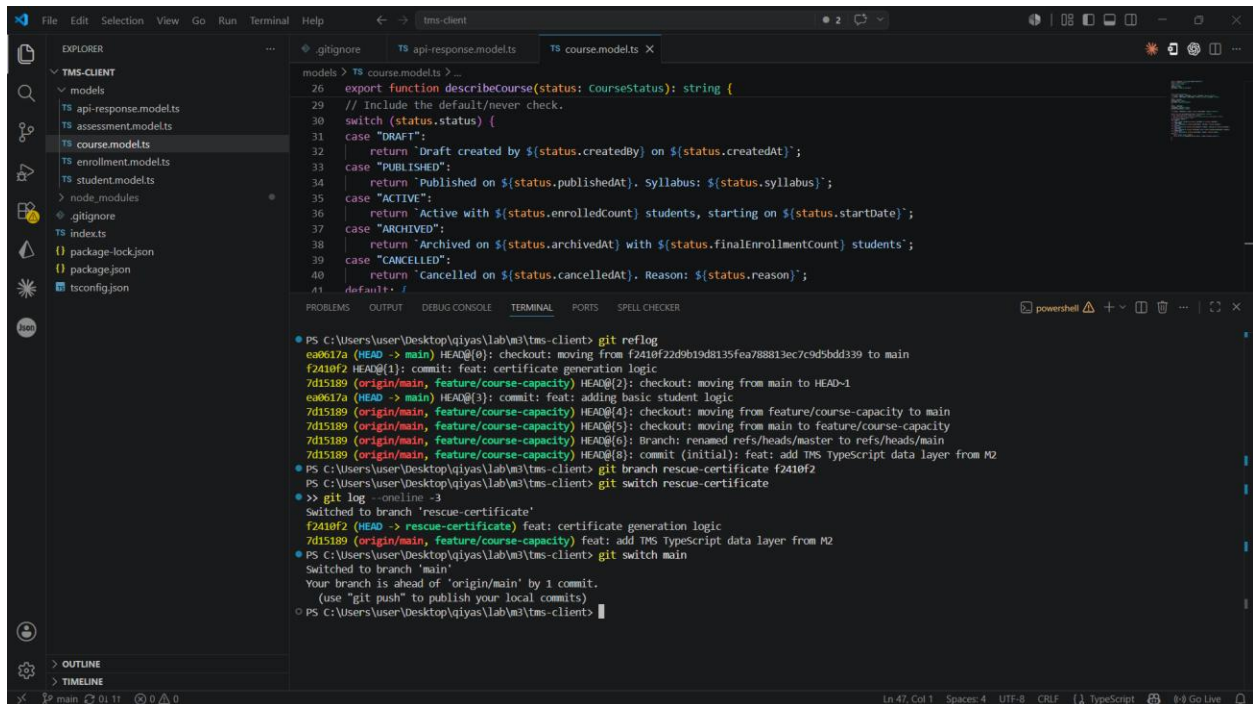
    f2410f2 feat: certificate generation logic

If you want to keep it by creating a new branch, this may be a good time
to do so with:

    git branch <new-branch-name> f2410f2

Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git log --oneline
ea617a (HEAD -> main) feat: adding basic student logic
7d15189 (origin/main, feature/course-capacity) feat: add TMS TypeScript data layer from M2
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Step 2 Recover with reflog



```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git reflog
ea617a (HEAD -> main) HEAD@{0}: checkout: moving from f2410f22d9b19d8135fea788813ec7c9d5bdd339 to main
f2410f2 HEAD@{1}: commit: feat: certificate generation logic
7d15189 (origin/main, feature/course-capacity) HEAD@{2}: checkout: moving from main to HEAD-1
ea617a (HEAD -> main) HEAD@{3}: commit: feat: adding basic student logic
7d15189 (origin/main, feature/course-capacity) HEAD@{4}: checkout: moving from feature/course-capacity to main
7d15189 (origin/main, feature/course-capacity) HEAD@{5}: checkout: moving from main to feature/course-capacity
7d15189 (origin/main, feature/course-capacity) HEAD@{6}: Branch: renamed refs/heads/master to refs/heads/main
7d15189 (origin/main, feature/course-capacity) HEAD@{8}: commit (initial): feat: add TMS TypeScript data layer from M2
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git branch rescue-certificate f2410f2
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch rescue-certificate
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git log --oneline -1
f2410f2 (HEAD -> rescue-certificate) feat: certificate generation logic
7d15189 (origin/main, feature/course-capacity) feat: add TMS TypeScript data layer from M2
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Exercise 3: Atomic Commit Literacy

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the file structure of a project named 'tms-client'. The file structure includes a 'models' directory with files like 'api-response.models.ts', 'assessment.models.ts', 'course.models.ts', 'enrollment.models.ts', 'student.models.ts', and 'EnrollmentValidator.ts'. The main editor area shows the 'EnrollmentValidator.ts' file with a single line of code: `1 Generate code (Ctrl+I), or select a Language (Ctrl+K M). Start typing to dismiss or don't show this again.`. The bottom panel shows the Terminal with the following commands and output:

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch -c feature/enrollment-capacity
Switched to a new branch 'feature/enrollment-capacity'
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git add models/enrollmentValidator.ts
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit -m "feat(enrollment): add max capacity validator"
[feature/enrollment-capacity 18afe9] feat(enrollment): add max capacity validator
1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 models/enrollmentValidator.ts
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Exercise 4: The Merge Conflict

Step 1

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the file structure of a project named 'tms-client'. The file structure includes a 'models' directory with files like 'api-response.models.ts', 'assessment.models.ts', 'course.models.ts', 'enrollment.models.ts', 'student.models.ts', and 'assessment-config.ts'. The main editor area shows the 'assessment-config.ts' file with two lines of code: `1 export const PASS_THRESHOLD = 50;` and `2`. The bottom panel shows the Terminal with the following commands and output:

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> echo "export const PASS_THRESHOLD = 60;" > assessment-config.ts
>> git add assessment-config.ts
>> git commit -m "chore: setup baseline assessment rules"
1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 assessment-config.ts
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch -c feature/strict-assessments
Switched to a new branch 'feature/strict-assessments'
>> echo "export const PASS_THRESHOLD = 75;" > assessment-config.ts
>> git commit -m "fix(assessment): increase threshold for honors certification"
[feature/strict-assessments 872ed3e] fix(assessment): increase threshold for honors certification
1 file changed, 0 insertions(+), 0 deletions(-)
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)
[main b6d13bc] fix(assessment): lower threshold for inclusive grading
1 file changed, 0 insertions(+), 0 deletions(-)
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git merge feature/strict-assessments
warning: Cannot merge binary files: assessment-config.ts (HEAD vs. feature/strict-assessments)
Auto-merging assessment-config.ts
CONFLICT (content): Merge conflict in assessment-config.ts
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Step 2

```
1 Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
2 export const PASS_THRESHOLD = 50;
3 =====
4 export const PASS_THRESHOLD = 75;
5 feature/strict-assessments (incoming change)
```

stage 3

```
1 export const PASS_THRESHOLD = 50; // This constant represents the minimum score required to pass an assessment.
2
3
```

```
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git add .
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit -m "(use the default merge message)"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git log --oneline --graph -5
* 842be39 (HEAD -> main, origin/main, feature/strict-assessments) assessment-config.ts
* ccf0751 fix(assessment): lower threshold for inclusive grading
|
* 872ed3e fix(assessment): increase threshold for honors certification
|
* b6d13bc fix(assessment): lower threshold for inclusive grading
|/
* 324fb01 chore: setup baseline assessment rules
PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
```

Exercise 5: The History Cleanup

Stape 1

File Edit Selection View Go Run Terminal Help

ts-client

EXPLORER

- TSMS-CLIENT
 - models
 - api-response.models.ts
 - assessment.models.ts
 - course.models.ts
 - enrollment.models.ts
 - student.models.ts
 - node_modules
 - .gitignore
 - assessment-configs.ts
 - index.ts
 - package-lock.json
 - package.json
 - tsconfig.json

assessment-configs.ts

```
1 export const PASS_THRESHOLD =50 ;// This constant represents the minimum score required to pass an assessment.
2
3
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

- PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit --allow-empty -m "chore: started certificate logic"
- [main 852199b] chore: started certificate logic
- PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>
- PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit --allow-empty -m "fix: forgot semicolon in PDF generator"
- [main bf5e5f4] fix: forgot semicolon in PDF generator
- PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit --allow-empty -m "oops: typo in student name"
- [main 27a341a] oops: typo in student name
- PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client> git commit --allow-empty -m "feat: finish PDF certificate logic"
- [main 90df01d] feat: finish PDF certificate logic
- PS C:\Users\user\Desktop\qiyas\lab\m3\tms-client>

main 28 minutes ago Ln 3, Col 1 Spaces: 4 UTF-16 LE CRLF TypeScript Go Live

Stape 2 and 3

[illegible]